

Course milestone M6: Data and Measurement Governance

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m06.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M06**. Brightspace carries the deadline.

This milestone presents **Book Milestone 6: Your data and measurement, governed** (checkpoint, version 1) of the course book, EDR|AI.

Open the milestone workbook in Colab

This chapter closes Studio 6: Govern data and measurement. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. Provenance documentation, a data-management record, your measurement specification, and a route-specific permission recheck.

What you bring

You also carry forward the last milestone's artifact: **Milestone 5: Your pathway, declared**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 20: Data Provenance and Data Quality: your acquisition route, written and dated, your provenance record for every dataset and borrowed number (the primary source behind your headline claim opened and read), your data-management update, and the permission recheck against the data as they actually arrived.
- Lesson 21: Measurement and Operationalization: your measurement specification: concept, construct, indicator, the meaning sentence for each number, the reliability check's result, the indicator-to-use validity argument with its strongest rival reading, and the boundary that follows.

The practice

1. Settle your acquisition route first, using the guidance above, and write down which route you took and why.
2. Document provenance for every source: who produced it, how, when, and under what terms.
3. Write your measurement specification: the construct, the operationalization, and the gap between them.
4. Assess reliability and validity for your own instrument, on the evidence you actually have.
5. Recheck permissions against the data as they actually arrived, not as planned.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Data that arrived differently from the plan can carry permissions the plan did not cover.
- **Evidence, provenance, and reproducibility.** Provenance is the evidence rail applied to your own data.

- **AI activity, verification, and human decisions.** An assistant can describe a dataset it has never seen; verify every claim against the file.
- **Uncertainty, claim boundary, and revision history.** Measurement error is uncertainty; state it alongside sampling uncertainty, not instead of it.

Where this milestone sits

You arrived carrying Milestone 5: Your pathway, declared. Milestone 7 runs the declared analysis on these governed data and produces your first reproducible result. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Any measurement change is a Contract version, because it changes what your estimand refers to.

How this milestone is assessed

Each row scores **0, 1, or 2. 10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: Provenance documentation, a data-management record, your measurement specification, and a route-specific permission recheck	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

! Blocking gate

No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.